Phase 4 Notes: Scaling PPO for the Discrete N-Ball Cilium Model

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1 Purpose of Phase 4

The goal of Phase 4 is to determine whether PPO can be trusted as a *scalable* optimizer for the discrete N-ball cilium stroke problem.

Earlier phases established exact and tabular methods for small systems. The current picture has two complementary tools. Howard average-reward policy iteration (Howard PI) provides an *exact* finite-MDP benchmark wherever the full transition/reward table is feasible to build. PPO provides a *sample-based* optimizer that never enumerates the full state graph and therefore does not inherit the table's curse of dimensionality. The central Phase 4 question is no longer "can PPO find a good stroke?" but

How close does PPO get to the exact Howard ceiling, and does that recovery stay reliable as N grows?

Because Howard PI gives the exact gain-optimal stroke for $N = 2, 3, 4$ at the discretizations we can table, we can answer this directly: run PPO seed sweeps, measure recovery of the Howard gain, and check whether PPO converges on one robust stroke family or scatters. If it behaves well where we can check it, we have earned the right to use it at $N = 5$, where Howard becomes infeasible.

Throughout, "optimal" means optimal for the finite deterministic MDP defined by the chosen angle grid, action set, reward, and boundary handling. These are notes about the discrete optimization landscape, not a continuum claim.

2 The generalized N-ball cilium environment

A state is the vector of relative joint angles $\phi = (\phi_1, \ldots, \phi_N)$ on a finite grid. The physical segment angles are cumulative,

$$\psi_k = \sum_{j=1}^{k} \phi_j, \qquad k = 1, \ldots, N,$$

so ϕ is the controlled state while ψ fixes the chain geometry. The default angle ranges are $\phi_1 \in [-\pi/4, \pi/4]$ and $\phi_k \in [-\pi/2, \pi/2]$ for $k \geq 2$. For $\Delta\theta = \pi/20$ this gives $n_{\text{bins}} = (11, 21), (11, 21, 21), (11, 21, 21, 21)$ for $N = 2, 3, 4$; the $N = 4$ grid has $11 \cdot 21^3 = 101{,}871$ states.

The action set is all nonzero vectors in $\{-1, 0, 1\}^N$, so there are $3^N - 1$ actions (80 for $N = 4$). Each nonzero component moves one angle by one bin. Boundary handling uses `clip_penalty` with invalid penalty -0.1 and reward rescaling 100. The reward is the one-step flux from the regularized Stokeslet/image solve, with wall-clearance and near-contact safeguards.

For exact methods the environment precomputes the MDP tables $R(s, a)$ (immediate reward) and $P(s, a) = s'$ (deterministic next state). These cached tables (the `vi_tables` files) are method-agnostic: value iteration, Howard PI, and cycle diagnostics all read the same R and P. For PPO the table is not built; trajectories are sampled with `precompute=False`. This is the core reason PPO scales: table methods cost states × actions, while PPO costs sampled environment steps. At $N = 4$ the table is ~ 8.1M state–actions (~ 23 min to build); at $N = 5$ it is ~ 0.5B (~ 25 GB, infeasible), which is the wall PPO is meant to step around.

3 Howard policy iteration as the exact benchmark

Howard policy iteration [2, 3] is the exact benchmark for the finite average-reward MDP defined by our discretized cilium model. In this setting, the objective is not discounted return over a finite or discounted horizon, but the long-run average reward of the periodic stroke.

The natural objective for a repeating stroke is the long-run average reward, or gain,

$$g^{\pi} = \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} R(s_t, \pi(s_t)).$$

In the deterministic finite setting, fixing a stationary policy turns the state space into a functional graph (one outgoing edge per state), so every trajectory is a transient path feeding into a periodic cycle. The gain is the mean reward around that cycle. Thus Howard PI solves precisely the discrete stroke-design problem: find a repeatable stroke of maximal average reward per step.

For a policy π the average-reward Bellman equation is [3, 1]

$$g + h(s) = R(s, \pi(s)) + h(P(s, \pi(s))),$$

with constant gain g on a recurrent class and bias h measuring transient preference. Howard PI alternates *evaluation* (identify the cycle under π, assign its mean reward as g, propagate h back along the transient trees) and *improvement* (at each s, pick the action maximizing $R(s, a) + h(P(s, a))$, prioritizing larger successor gain). A sticky-tie rule keeps the current action when it is optimal within tolerance, preventing chatter among exact ties. It terminates when no state changes action, at which point π is gain-optimal; rolling out from the midpoint state yields the cycle start, length, and mean reward.

This differs from discounted value iteration, $V_\gamma(s) = \max_a[R(s, a) + \gamma V_\gamma(P(s, a))]$. As $\gamma \to 1$ the discounted optimum approaches the average-reward optimum, but V_γ carries a large common component $\approx g/(1-\gamma)$. Howard PI targets the gain directly and needs no discount factor, making it the natural exact benchmark for a periodic stroke. (This also resolves an earlier puzzle: learned methods that "beat VI" on average reward were closer to the *gain*-optimal cycle, which discounted VI does not target. Howard PI is not beatable — it is the finite-MDP ceiling.)

Howard benchmarks. Table 1 lists the gain-optimal cycles. The per-step gain is roughly linear in N (≈ 0.4 per added link). The cycle *geometry* is not monotone: at $\Delta\theta = \pi/20$, the $N = 3$ optimum is a doubled (two-pass) orbit of length 48, whereas $N = 2$ and $N = 4$ are single strokes (Section 6.2 explains how this is identified).

4 PPO as the scalable method

PPO is the object we hope to scale [4, 5]. The observation is the normalized angle-bin state; *no* curvature, smoothness, or phase features are supplied. That makes the results notable: PPO finds coherent strokes directly from the raw discrete representation. The headline metric is

$$\text{recovery} = \frac{\text{PPO mean cycle reward}}{\text{Howard gain}},$$

N	$\Delta\theta$	Howard gain g	Howard L	note
2	$\pi/20$	0.3989	25	single
2	$\pi/30$	0.2659	38	single
3	$\pi/20$	0.8014	48	doubled ($\approx 2 \times 24$)
3	$\pi/30$	0.5394	35	single
4	$\pi/20$	1.2399	21	single
4	$\pi/30$	0.8313	33	single

Table 1: Howard gain-optimal benchmarks. Per-step gain drops at $\pi/30$ because each action is a smaller angular step (see Section 7); the single-pass flux per stroke is resolution-invariant.

computed at matched N, $\Delta\theta$, reward, and boundary settings, and using the same cycle-extraction (roll out the deterministic policy from the midpoint, detect the first repeated state). The matched-evaluation point matters: PPO staying below 100% is a sanity check *only* if PPO and Howard share the reward convention and extraction, which they do here (both read the same R). A reported $> 100\%$ would first implicate an evaluation mismatch, not a super-optimal stroke.

Protocol (identical for each N). Fix $\Delta\theta$; train PPO for 10^6 steps; run seeds 0–9; roll out the greedy policy from the midpoint; detect the cycle; record length, mean cycle reward, and percent of Howard gain. We additionally log three geometric diagnostics (Section 6) that need no Howard reference.

5 Sweep results: $N = 2, 3, 4$ at $\pi/20$ and $\pi/30$

All runs are 10^6 steps, 10 seeds each. Table 2 summarizes recovery; the per-seed cycle lengths are listed to expose the doubled outliers.

N	$\Delta\theta$	g	L_H	median %	mean %	min %	max %	lengths
2	$\pi/20$	0.3989	25	88.0	42.2	-275.8	94.4	17,18,21,22,24,25
2	$\pi/30$	0.2659	38	82.3	67.9	-32.7	85.7	22,25,30,31,32,33
3	$\pi/20$	0.8014	48	93.9	93.4	90.2	96.3	19,20,21,22,23
3	$\pi/30$	0.5394	35	89.0	87.2	74.0	94.9	24,26,28,30,31,32
4	$\pi/20$	1.2399	21	94.4	93.1	87.6	95.9	18,19,20,38
4	$\pi/30$	0.8313	33	89.9	89.9	86.3	94.6	26,27,28,31,56

Table 2: PPO recovery of the Howard gain over 10 seeds. "%" is percent of Howard gain. The mean–median gap at $N = 2$ is driven by degenerate seeds (Section 6.1); at $N \geq 3$ mean and median agree to ~ 1 point.

Figure 1 summarizes the main PPO–Howard comparison across N. The points show the median recovery over the 10 full 10^6-step PPO seeds, while the whiskers show the full minimum–maximum range across seeds. The typical PPO run improves from $N = 2$ to $N = 3, 4$, recovering roughly 90–96% of the Howard gain for $N = 3, 4$. The finer $\Delta\theta = \pi/30$ grid is consistently a little harder than $\Delta\theta = \pi/20$, but the drop is modest and does not change the qualitative picture. The large lower whiskers at $N = 2$ reflect the wrong-orientation failure modes discussed below; these are clipped in the zoomed view so that the behavior of the typical and high-performing seeds is visible.

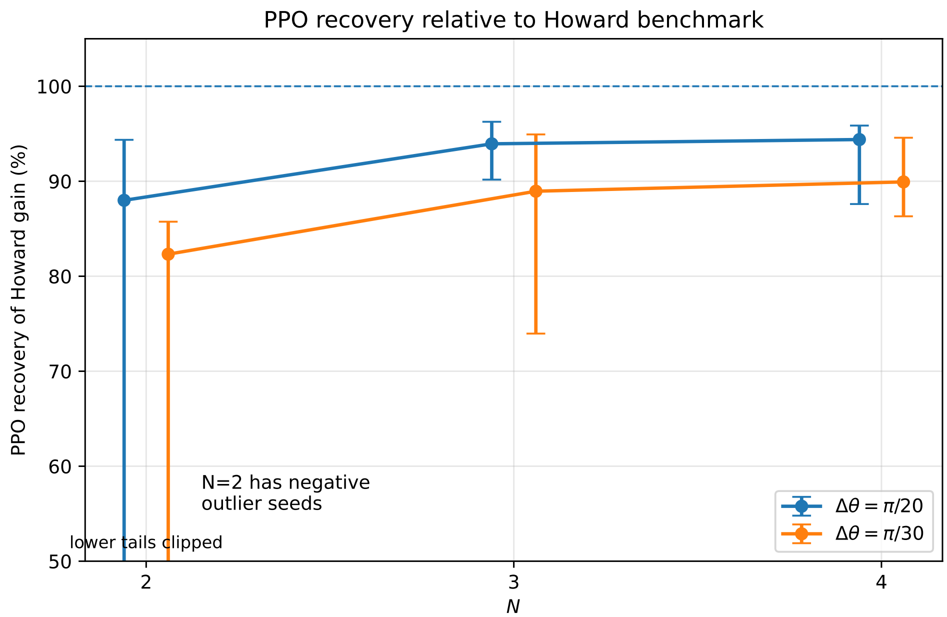

Figure 1: PPO recovery relative to the Howard gain benchmark. Points show the median recovery over 10 full 10^6-step PPO seeds; whiskers show the minimum–maximum range. The zoomed view clips the negative $N = 2$ outliers, which are diagnosed separately as wrong-orientation pumping failures.

Reliability improves with N. The central empirical result is that the *typical* (median) seed recovers $\sim$ 88–94% at $\pi/20$ and $\sim$ 82–90% at $\pi/30$, and that reliability *increases* from $N = 2$ to $N = 4$. At $N = 3$ and $N = 4$ all ten seeds land in a tight 87–96% band with no degenerate runs; only $N = 2$ produces failures. This is the opposite of the naive scaling worry and the most encouraging single fact for $N = 5$: the small problem is the dangerous one, not the large one. (The earlier 100k-step $N = 2$ runs sat at 50–54%, confirming that the full 10^6 steps are needed for the recovery numbers above.)

One stroke family. Across best, median, and long-cycle seeds, and across both resolutions, the strokes are the same power-stroke/recovery family: the same single-loop tip path, the same fan in the overlay, the same positive-burst/negative-trough reward signature. PPO is not scattering across qualitatively distinct gaits — it is converging on one geometric stroke and recovering $\sim$ 90% of its optimal execution. This cross-seed, cross-resolution *continuity* is the property we will lean on at $N = 5$ once the Howard ceiling is gone.

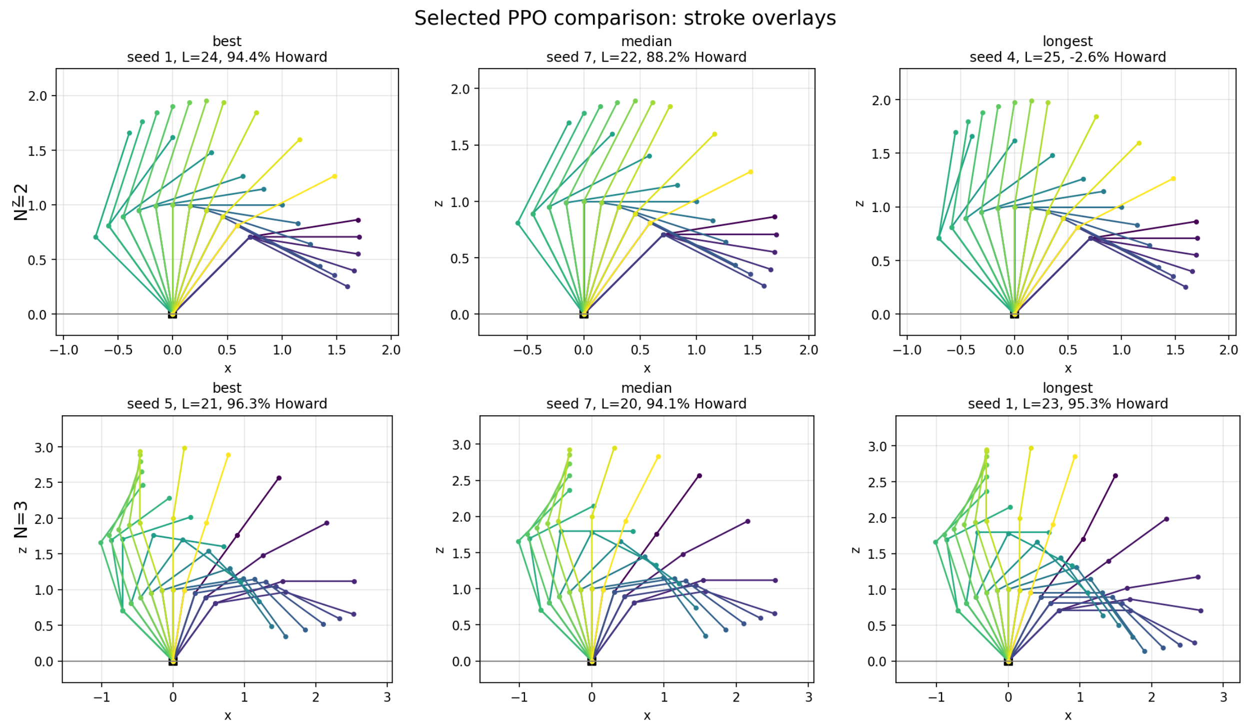

Figure 2: $N = 2$ (top) and $N = 3$ (bottom) PPO stroke overlays: best, median, and longest seed. The gait is consistent across seeds; the $N = 2$ "longest" panel is a degenerate seed (Section 6.1).

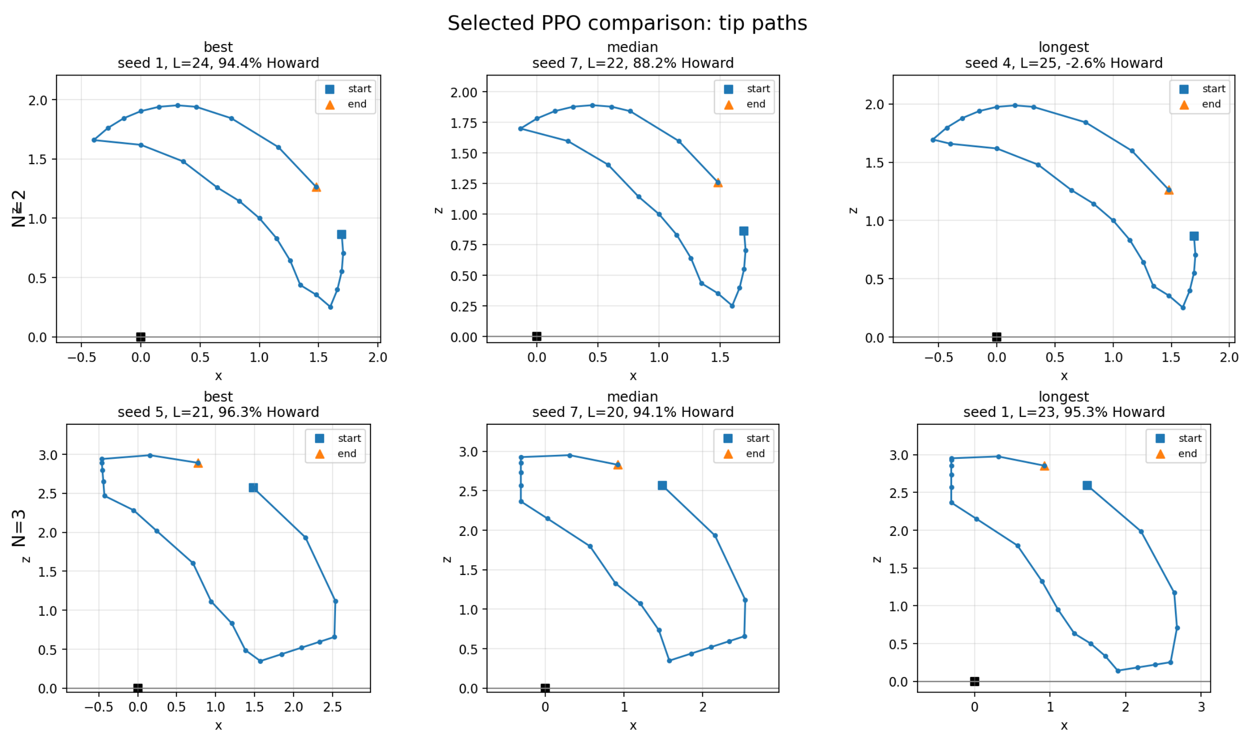

Figure 3: $N = 2/N = 3$ PPO tip paths. Loop area grows with N; the enclosed area is the pumping (Section 6).

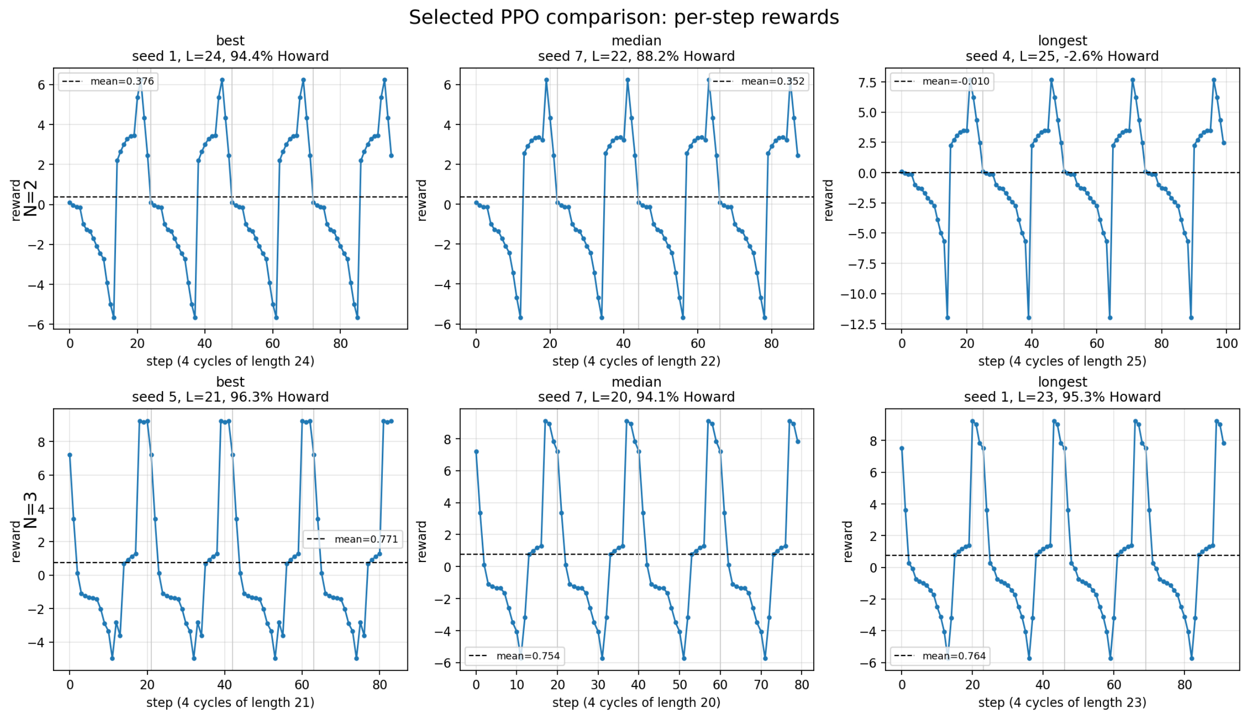

Figure 4: $N = 2/N = 3$ PPO per-step reward traces. Note the negative-mean degenerate $N = 2$ seed (positive-looking geometry, backward pumping).

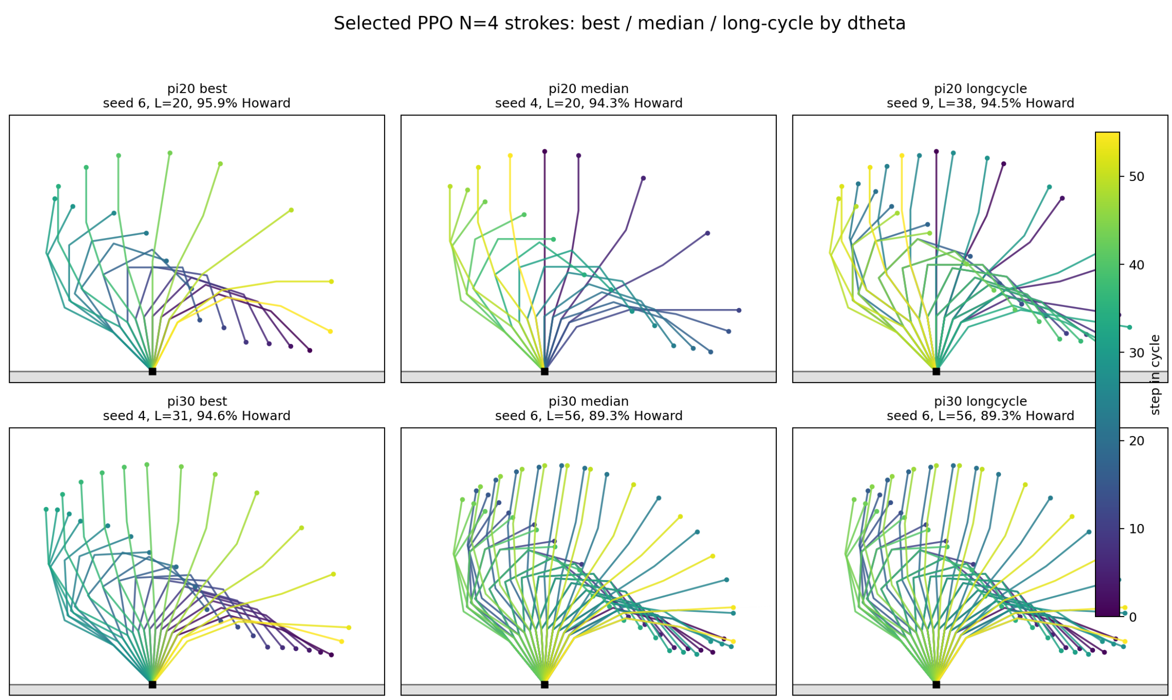

Figure 5: $N = 4$ PPO stroke overlays, best/median/long-cycle at $\pi/20$ and $\pi/30$. The long-cycle panels ($L = 38$, $L = 56$) are doubled orbits (Section 6.2).

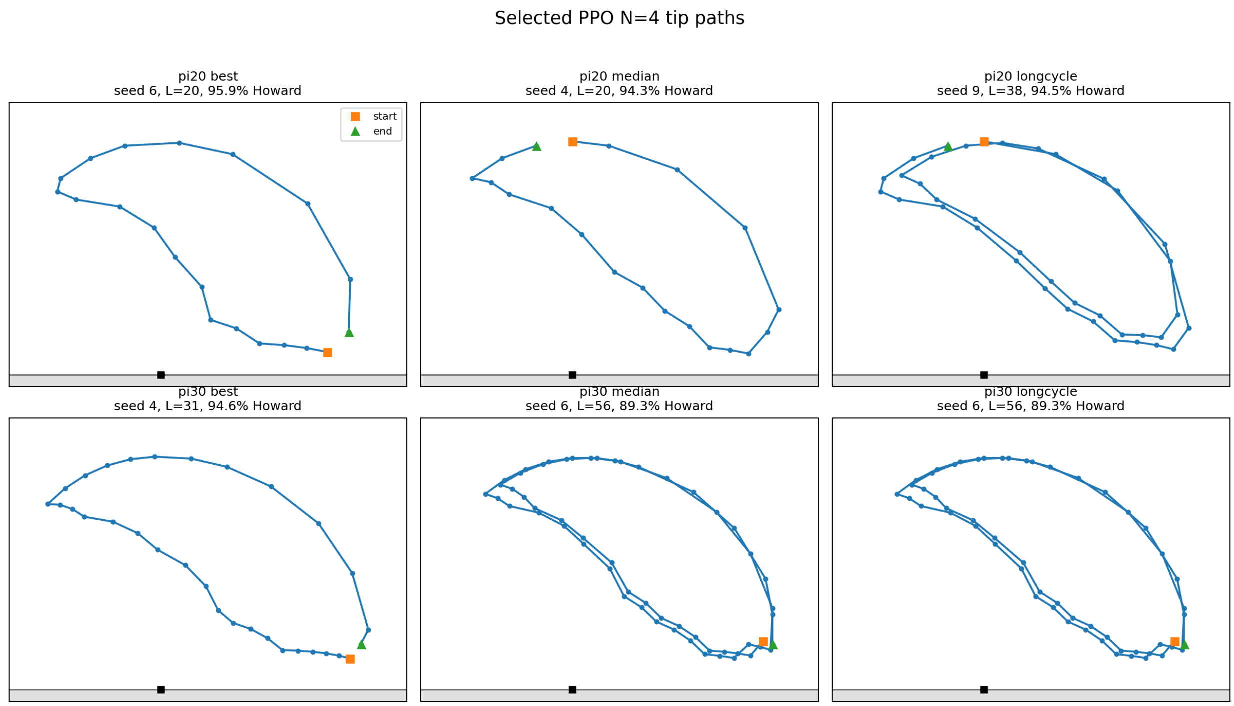

Figure 6: $N = 4$ PPO tip paths. The doubled-orbit panels trace the loop twice ($\sim 2\times$ path length and area).

6 Geometric diagnostics (Howard-free)

Beyond recovery, we log three quantities per stroke that require no Howard reference: the enclosed tip-path area A, the tip-path length ℓ, and the ratio $\rho = (\text{cycle reward})/A$. These turn out to carry most of the physics and to classify every failure mode we have seen.

Net pumping tracks enclosed tip area. Empirically, ρ is nearly constant *within each* N, across good seeds, degenerate seeds (in magnitude), doubled orbits, and both resolutions (Table 3). This is consistent with a geometric-phase / Green's-theorem picture for a low-Reynolds swimmer, in which net pumping over a cycle scales with the area enclosed by the stroke in configuration/tip space rather than with the path traversed. The key practical consequence: ρ is a *physics-grounded, benchmark-free* gait-quality metric, and A itself rises monotonically with N (Table 3), exactly as the rising pumping requires.

N	median tip area A ($\pi/20$, $\pi/30$)	$\rho =$ cycle reward$/A$
2	0.27, 0.22	≈ 31
3	0.44, 0.39	≈ 36
4	0.54, 0.49	≈ 42

Table 3: Tip-area grows with N; reward-per-area ρ is nearly invariant within N (forward-pumping seeds). Finer grid gives slightly smaller area (same gait, executed a touch less fully).

6.1 Failure mode 1: wrong-orientation collapse

The $N = 2$ degenerate seeds (e.g. $\pi/20$ seeds 4 and 6, $\pi/30$ seed 7) are *not* zero-area reciprocal collapses. Their tip area is normal-to-large, but ρ is *negative*: the chain sweeps a healthy loop but traverses it in the wrong orientation, pumping backward. The angle waveforms look like the good gait; only the reward (or the sign of ρ) exposes them. This is a more interesting failure than "did not learn" — it learned the stroke and runs it the wrong way around the loop. It appears only at $N = 2$ (roughly 2–3 of 10 seeds), and never at $N \geq 3$ in these sweeps. *Diagnostic: $\rho < 0$ flags it with no Howard reference.*

6.2 Failure mode 2: commensurate doubling

A few cycles are doubled (two-pass) orbits: the stroke repeats nearly twice before closing on the discrete grid, because the physical period does not commensurate with the bin spacing. These are now cleanly identified, settling the verification the analysis called for:

- **Path-length / area signature.** A doubled orbit has ℓ and A roughly *double* the in-family value while ρ stays in band. The two PPO doubled seeds — $N = 4, \pi/20$ seed 9 ($L = 38$) and $N = 4, \pi/30$ seed 6 ($L = 56$) — both show $\ell \approx 6.3$ and $A \approx 1.0$ versus $\ell \approx 3.2$, $A \approx 0.5$ for neighbors, with $\rho \approx 42$ unchanged. That constant ρ across the doubling *is* the proof the long orbit is two copies of the same gait, not a new one.
- **Flux-per-stroke (single-pass).** $g \times L$, halved for doubled orbits, is resolution-invariant (Table 4). This is also how the Howard references are labelled single vs. doubled: only $N = 3, \pi/20$ ($L = 48$) is doubled on the Howard side (its $g \times L = 38.5$ is $2\times$ the $\pi/30$ single-pass 18.9); $N = 2, \pi/30$ ($L = 38$) is a *single* stroke at the finer grid, not a doubled one.

Doubling does *not* improve reward relative to the best shorter-cycle strokes — it is neutral-to-worse — so it is a rare curiosity, not a competing gait. *Diagnostic: ℓ (or L) $\approx 2\times$ the in-family value flags it.*

N	$\Delta\theta$	g	L_H	$g \times L_H$	single-pass flux
2	$\pi/20$	0.3989	25	9.97	9.97
2	$\pi/30$	0.2659	38	10.10	10.10
3	$\pi/20$	0.8014	48	38.47	19.24 (doubled)
3	$\pi/30$	0.5394	35	18.88	18.88
4	$\pi/20$	1.2399	21	26.04	26.04
4	$\pi/30$	0.8313	33	27.43	27.43

Table 4: Single-pass flux per stroke is resolution-invariant ($\sim 10, \sim 19, \sim 26$ for $N = 2, 3, 4$) once the doubled $N = 3, \pi/20$ value is halved. This is the correct cross-$\Delta\theta$ quantity; per-step gain is not, because it scales with the angular step.

7 Cross-resolution comparison

The finer grid costs a bounded $\sim$ 4–6 median points at every N ($88 \to 82$, $94 \to 89$, $94 \to 90$) and widens the spread modestly — a sample-efficiency tax, not a qualitative break. Per-step Howard gain drops by $\approx 20/30$ at $\pi/30$ (Table 1) because each action is a smaller angular displacement;

with dt fixed, the same physical stroke is executed in more, smaller steps. The resolution-invariant quantity is therefore single-pass flux per stroke (Table 4), and the percent-of-Howard normalization is fair because each grid is compared to its own ceiling.

8 Interpretation and scale-up plan

The Phase 4 story is now well supported. Howard PI gives the exact finite-MDP ceiling where tables are feasible; PPO approaches it from below, recovering $\sim$ 90–96% at $N = 3, 4$ without enumerating the state graph. More importantly, PPO finds a stable geometric stroke family across seeds and resolutions, and reliability *improves* with N over $N = 2, 3, 4$. The two known failure modes are both rare and both detectable without a benchmark: wrong-orientation collapse ($\rho < 0$, only at $N = 2$) and commensurate doubling ($\ell \approx 2\times$ in-family, rare at $N = 4$).

This is the validation needed to move past the tabular wall. At $N = 5$ no Howard ceiling exists, so confidence will rest on three benchmark-free signals, all exercised above: (i) cross-seed agreement on one stroke family; (ii) continuity — the $N = 5$ stroke should be a natural continuation of the $N = 2, 3, 4$ family (rising tip area, rising ρ, single-loop non-reciprocal tip path); and (iii) the diagnostic audit — flag $\rho < 0$ (backward pumping) and $\ell \approx 2\times$ (doubling) automatically across the sweep.

Next steps.

1. Run $N = 5$ at $\Delta\theta = \pi/20$ with the validated PPO workflow and the benchmark-free audit: cross-seed agreement, tip area, tip-path length, reward-per-area, and checks for $\rho < 0$ or doubled paths.

2. Test whether a factored `MultiDiscrete` action representation improves scaling. The current monolithic action head grows as $3^N - 1$; a factored ternary action has one head per joint and grows linearly with N. Revalidate at $N = 3, 4$ before relying on it for larger N.

3. Decide whether curvature-aware or smoother state features are necessary only after seeing the $N = 5$ sweep. The raw angle-bin observation already gives a coherent stroke through $N = 4$, so this is a scale-up question rather than a prerequisite.

Working conclusion.

> For the generalized N-ball cilium environment, Howard policy iteration[2] gives an exact average-reward benchmark for small N, and PPO recovers a large fraction of it without enumerating the state graph. Across $N = 2, 3, 4$ and two resolutions, PPO converges on a single coherent stroke family, with recovery $\sim$ 90–96% at $N = 3, 4$ and reliability improving with N. Two rare failure modes — backward-pumping (wrong loop orientation) and commensurate doubling — are both detectable from benchmark-free reward–geometry diagnostics: the reward-per-tip-area sign and the tip-path-length doubling. These benchmark-free diagnostics, together with cross-seed family agreement and continuity across N, support using PPO as the primary optimizer at $N = 5$ and beyond, where Howard PI is infeasible.

References

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